Respiratory-Phase Multipath Nulls in mmWave Cardiac Monitoring: Depth Bound, Saturation, and a Deployable Criterion

XinHao Yu

***Abstract*—This article treats the deep nulls that interrupt mmWave cardiac radar as a propagation event with a computable onset, and shows that range resolution is the design variable that moves their incidence. A bin that sums a chest return with a static return collapses when the two phasors reach antiphase; existing bin-selection and clutter-removal methods act after the fact and never state when a collapse is possible. We derive the null condition and a closed-form depth bound whose minimum lies at $\rho = 1$. Beyond half a turn the bound loses its swing dependence and reduces to $|\rho-1|/(1+\rho)$—which turns a two-condition failure into a single band $\rho \in [0.7391, 1.3529]$; breathing at 60 GHz is past that point in 76.4% of our records. The method has two parts: a gate, the fraction of frames below a depth threshold, and a design rule, the chirp bandwidth that places ρ outside the band. In a dual-subject 60 GHz dataset with the geometry held fixed, the share of records carrying at least five nulls rises from 13.0% to 62.3% as bins coarsen from 5.0 to 7.5 cm, the gate predicts the second half of a record from the first (AUC 0.91, 0.89 held out by position), and the closed form reproduces 1450 individual depths to 0.057. Unlike an amplitude fade, which leaves the same envelope statistics, the measured dips carry a phase step that grows with depth, separating the two mechanisms at an AUC of 0.71. Breath-hold in 110 subjects removes 56% of the collapses with motion unchanged, so the null is respiratory in origin and avoidable by design.**

***Index Terms*—mmWave radar, contactless physiological monitoring, multipath destructive interference, deep fade, range-bin selection, phase unwrapping, range resolution.**

I. Introduction

THIS article answers one question: when does a range bin in mmWave cardiac radar collapse, and what does a designer change to prevent it? The prevailing assumption is that the phase of one bin tracks chest-wall displacement alone. Indoors it does not: desktops, walls, the radar mount and the body share the bin with the chest, and when the two returns rotate into antiphase the amplitude collapses, the unwrapped phase jumps, and the downstream chain fails as a unit. The literature treats this as a selection problem—rank the bins better, or remove the static return—and both families work after the fact.

We take a different view: the collapse is an event with an onset condition, and that condition is a ratio the designer already controls. Because the fade is caused by how much static power shares the bin, and the bin width is $\Delta R = c/2B$, the chirp bandwidth is not merely a resolution setting; it is the parameter that decides whether a null is possible at all. The relevant literature splits along that line. Multiple-bin fusion [1], [2], sub-range-bin quality indexes [3], temporal phase coherency [4], and array-side cardiac beamforming [5] all improve the odds of picking a usable bin, but they rank bins after the fact; moving-target indication [6], static-return rejection [7] and quadrature calibration [8] remove the static return without asking when the remainder will null. Neither family connects the collapse to a bandwidth the designer can change.

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This paper supplies that connection in three steps. First, a two-phasor model gives the exact null condition, a closed-form depth bound, and the swing and displacement thresholds implied by any depth criterion. Second, we show that the amplitude ratio ρ is the quantity that range resolution moves, and test this on raw ADC from a 60 GHz radar where the subjects and the nine geometries are held fixed while the chirp bandwidth is swept. Third, we give an online statistic D that flags a bin before it is used, plus the reference-frame condition under which a null can be assigned a fixed respiratory phase.

Design implications. The results argue against a single "best" resolution. A fine bin isolates the chest and becomes static-dominated ($\rho \ll 1$), which suppresses nulls but also suppresses the displacement signature; a coarse bin mixes the chest with room clutter and moves ρ toward one, where nulls appear. The usable operating point is a band in ρ, not an extreme, and for the four fifths of records whose swing is already past saturation, that band is the whole of the model: what a designer controls is ρ, and ρ is the bin width. Four rules follow, and a scoring penalty Q_D applies them to bin ranking.

Three terms, kept apart. A *mathematical null* is the exact cancellation $\rho = 1$, $\theta = \pi$, a property of the ideal signal; a *deep dip* is a local minimum of $|z|$ below $0.35\times$ the record median, which is what a detector sees; a *failure event* is five or more such dips in 30 s (R1), which is what a system acts on. The closed form is derived for the first, the statistics are counted on the second, and the gate is calibrated on the third. An envelope dip is not by itself a geometric null: the latter is a dip the alignment and shape statistics attribute to cancellation, a claim about mechanism rather than about the envelope.

Contributions.

1) *Mechanism.* A two-phasor formulation with a closed-form depth bound, its single minimum at $\rho^* = 1$, the saturation above π that leaves a band in ρ alone, and the translation of a depth criterion into hardware numbers (15% $\Leftrightarrow$ 2.84 rad $\Leftrightarrow$ 2.26 mm at 60 GHz).

2) *Validation.* A bandwidth-controlled experiment on raw ADC from a dual-subject 60 GHz radar (13.0% to 62.3% across 5.0 to 7.5 cm), an event-level test of the closed form over 1450 dips to 0.057 in depth, a shape test that separates cancellation from amplitude fading against two depth-matched synthetic worlds, and a breath-hold intervention in 110 subjects (56% of collapses removed, motion unchanged).
3) *Engineering criterion.* A gate of one count and one threshold that predicts the second half of a record from its first (0.91 pooled, 0.89 held out by position) and keeps its ordering on a second front end ($+0.58$), with rules R1–R4 and their scope.

II. Related Work

Radar vital-sign chains and range-bin selection. FMCW radar separates reflectors in range and reads the phase of one or several bins. Recent work pushes the chain toward robust low-SNR cardiac monitoring: a 4-D cardiac beamformer focuses array beams on the heart [5], and binary phase modulation separates multiple subjects' thoracic micro-motion [9]. Within bin selection, Szmola *et al.* [1] reported that multi-bin fusion cuts heart-rate error from 1.97 to 0.66 bpm and later applied it to acute stroke patients [2]; Choi *et al.* exploited temporal phase coherency [4]; Yuan *et al.* proposed sub-range-bin quality indexes [3]. Phase integrity is a standing concern, addressed by quadrature-imbalance calibration [8]. Gruzewska *et al.* [10] fed 32 range cells to a transfer-learning pipeline and noted the risk of averaging adjacent cells with equal frequency but opposite phase; Zhang *et al.* [11] achieved training-free long-term monitoring via voxel selection and high-order harmonics. These works establish that bin selection dominates signal quality, yet they treat a failing bin as a selection error to be avoided, not as a propagation event with a computable onset condition.

Multipath, null points and deep fades. Cancellation of two coherent reflections is classical in indoor radio propagation [12]: when the phasors are comparable in magnitude and in antiphase, the received amplitude falls toward the noise floor. In radar vital signs the second phasor is not static but rotates with the 0.2–0.5 Hz respiratory phase, so the fade is periodic rather than spatial, which is what makes the failure look like an algorithm dropout rather than a propagation outage. The phenomenon is named in this field—the *null point*—and is treated as a defect to be engineered away: frequency tracking shifts the carrier off the null [13]; phase- and self-injection locking with path diversity keeps one branch off null [14]; dual null detection points are removed with their time-domain sensitivity analysed [15]; differential quadrature demodulation extends tolerance to a dynamic offset [16]; and rejecting the static return is the alternative route [6], [7]. All share one premise: the null can be avoided, cancelled or calibrated out. None reports how deep the residual null remains once the removal is imperfect—and that residual decides whether the downstream chain survives. The field's three standing complaints—null point, DC offset and nonlinear distortion—are one condition at $\rho \to 1$ seen from three angles, and the depth of that condition is what this paper bounds.

Datasets and benchmarks. The released datasets span very different range resolutions: 31.25 cm [17], 4.5 cm [18], 5.0–7.5 cm [19] and an unreleased capsule [20]. Only [19] holds the subjects and geometries fixed while sweeping bandwidth on raw ADC, which is why we use it as the resolution-controlled block; the other datasets supply the causal intervention [17], the phase analysis [18], [20], and a cross-hardware check [18].

III. Formulation: Two Phasors and a Depth Bound

This section derives the three quantities the rest of the paper uses: a null condition, a closed-form depth bound, and the amplitude ratio on which both depend.

A range bin sums every reflector within its width, and two properties of that sum fix the minimal model. The sum is dominated by whatever does not move, because static returns add coherently and never average out; the only part that moves is the chest, whose displacement enters through the carrier phase. One static phasor plus one motion phasor is therefore the least structure the problem admits, not an assumption about the room. We state each consequence below in a form a measurement can refute, and we test the mechanism rather than fit individual samples: the statistics that carry the argument are rates, depths and correlations over a population.

A. Two-Phasor Signal

The bin signal (1) is the sum of the two,

$$z(t) = A_s + A_c\, e^{j\theta(t)}, \qquad \theta(t) = \frac{4\pi d(t)}{\lambda}, \tag{1}$$

where $A_s > 0$ is the static phasor (the frame is rotated so that A_s is real), $A_c \geq 0$ is the motion-phasor amplitude, $d(t)$ is the radial chest displacement and λ is the carrier wavelength. Expanding gives

$$|z|^2 = A_s^2 + A_c^2 + 2A_sA_c\cos\theta, \quad \frac{|z|}{A_s} = \sqrt{1+\rho^2+2\rho\cos\theta}, \tag{2}$$

with $\rho = A_c/A_s$. Fig. 1(a) draws that trajectory: a circle of radius A_c centred at A_s, which reaches the origin only when $\rho = 1$.

B. Null Condition

The bin nulls when the two phasors cancel, which requires $\rho = 1$ and $\theta = \pi \pmod{2\pi}$: only then do the two terms of (2) vanish together. A null is therefore a ratio condition and a phase condition at once, and a subject whose swing never reaches antiphase cannot produce one, however favourable the ratio. If $\theta_{\max} > \pi$ the trajectory crosses the antiphase direction twice per respiratory cycle.

C. General Depth Bound

For a swing $\theta \in [-\theta_{\max}, \theta_{\max}]$, $\cos\theta$ is minimized at the endpoints, so the deepest point of the arc fixes the ratio. Hence

$$r(\rho, \theta_{\max}) = \frac{\sqrt{1+\rho^2+2\rho\cos\theta_{\max}}}{1+\rho}, \tag{3}$$

where $r = |z|_{\min}/|z|_{\max}$ and $|z|_{\max} = A_s + A_c$ at $\theta = 0$. The bound assumes a swing symmetric about the static phasor, $\theta \in [-\theta_{\max}, \theta_{\max}]$, and takes the minimum over that arc. Because that minimum moves as the arc grows, the bound has two branches. While $\theta_{\max} \le \pi$ the deepest point is an endpoint and the swing enters through $\cos\theta_{\max}$. Once the arc covers the antiphase direction the deepest point is fixed at $\theta = \pi$, and (5) replaces (3) rather than extending it. An arc centred at $\theta_0 \neq 0$ moves the onset but not the saturated branch, because a half-width $\theta_{\max} \ge \pi$ reaches the antiphase direction for every θ_0.

Differentiating r^2 with respect to ρ gives a numerator proportional to $(1 - \cos\theta_{\max})(\rho - 1)$, so the bound has its single minimum at $\rho^* = 1$ and weakens monotonically on either side. Substituting,

$$r(1, \theta_{\max}) = \sqrt{\frac{2 + 2\cos\theta_{\max}}{4}} = \left|\cos\frac{\theta_{\max}}{2}\right|, \qquad \theta_{\max} \le \pi, \tag{4}$$

and $r = 0$ once $\theta_{\max} \ge \pi$.

Saturation: above half a turn the swing drops out. Once $\theta_{\max} \ge \pi$ the arc encircles the origin, the deepest point is always $\theta = \pi$, and (4) loses its dependence on the swing entirely: the bound reduces to

$$r = \frac{|\rho - 1|}{1 + \rho}, \qquad \theta_{\max} \ge \pi, \tag{5}$$

so the 15% criterion becomes the band $\rho \in [0.7391, 1.3529]$ and nothing else—the two roots of $|\rho - 1|/(1 + \rho) = 0.15$, that is $0.85/1.15$ and $1.15/0.85$. In this regime the criterion is one-dimensional, and the only quantity left to design against is the amplitude ratio.

Real records reach that regime. At 60 GHz the measured peak swing has median 5.92 rad and exceeds π in 76.4% of our records (Fig. 2(b)), so the swing threshold acts as an *onset* condition rather than a second design knob: it states how shallow a breath must be for a deep null to be impossible, and it governs only the one fifth of records that breathe too weakly to reach the saturated branch. We estimate $\theta_{\max}$ as the angle of z about its fitted circle centre, which equals $\theta(t)$ in the ideal model and costs a circle-fit residual of 0.321 of the radius in measured data (Section VI-B). Because that angle is taken about the circle's own centre, it is the motion parameter rather than the phase about the origin, and it is not compressed at large swing; the 76.4% figure is a direct measurement.

From a depth criterion to hardware numbers. Inverting (4) turns any chosen depth criterion into a required swing. For the 15% hard-failure line, $\theta_{\max} = 2\arccos 0.15 = 2.840$ rad. Since $\theta(t) = 4\pi d(t)/\lambda$, a phase semi-amplitude $\theta_{\max}$ corresponds to a peak-to-peak chest displacement

$$a_{pp} = \frac{\theta_{\max}\lambda}{2\pi}. \tag{6}$$

At 60 GHz ($\lambda = 5$ mm) the 15% criterion is therefore reached at $a_{pp} = 2.26$ mm, and at 77 GHz it would be reached at 1.76 mm. This is the origin of rule R4: a subject breathing shallower than this cannot produce a 15% null at $\rho = 1$ regardless of the room, so the bin is exempt. The number is a property of the carrier and the criterion, not of the subject pool. R4 covers the 20.5% of records whose breathing is too shallow to reach the onset; the other four fifths sit in the saturated regime of (5), where the criterion is one-dimensional.

D. Small-Signal Compression and Ratio Inversion

For small swings the static phasor compresses the observed phase, which is why the ratio cannot be read off the phase alone. With $\phi = \arg z$,

$$\left.\frac{d\phi}{d\theta}\right|_{\theta=0} = \frac{A_c}{A_s + A_c} = \frac{\rho}{1 + \rho} \equiv c(\rho). \tag{7}$$

Writing $x = \Delta\phi_{\text{app}}/\Delta\theta_{\text{true}} = c(\rho)$, inversion gives

$$\hat{\rho} = \frac{x}{1 - x}, \qquad \Delta\theta_{\text{true}} = \frac{4\pi\, a_{pp}}{\lambda}, \tag{8}$$

(8) recovers the ratio from the measured phase compression, with a_{pp} the peak-to-peak chest displacement.

Two limits belong to (7): it is a small-signal slope, and it describes the phase about the origin, which saturates at $2\arcsin\rho$ ($\rho < 1$) or $\theta_{\max}/2$ ($\rho = 1$) once the swing passes π. This paper therefore takes ρ from amplitudes and $\theta_{\max}$ about the fitted centre, where neither compression nor saturation enters.

E. Amplitude Estimate of the Ratio

Amplitudes give the ratio directly and survive that compression, which is what makes the criterion usable on a compressed phase. With $a[n] = z[n] - \hat{A}_s$,

$$\hat{A}_s = \left|\frac{1}{N}\sum\nolimits_n z[n]\right|, \quad \hat{A}_c = \sqrt{\frac{1}{N}\sum\nolimits_n |a[n]|^2}, \quad \hat{\rho}_A = \frac{\hat{A}_c}{\hat{A}_s}. \tag{9}$$

The RMS form is used because $a(t)$ is a rotating phasor: $|a| = A_c$ at every instant, whereas the range of $|a|$ would be zero. In practice (9) fixes the order of magnitude. Throughout this paper the operating statistic is $|A_s|/R$, with R the fitted motion-circle radius; this is the reciprocal of ρ up to the difference between $\hat{A}_c$ (an RMS) and R (a circle radius), and every statement below stays in the variable that is actually measured.

The estimator carries a bias that is bounded and known. For a symmetric swing $\frac{1}{N}\sum_n z[n] = A_s + A_c J_0(\theta_{\max})e^{j\theta_0}$, so the relative bias is $\rho|J_0(\theta_{\max})|$. At the measured median swing of 5.92 rad ($|J_0| = 0.205$) it is 1.4%, 5.4% and 7.7% at the three bandwidths; over all 161 records it is 2.2% for the 128 outside the failure band and 26% for the 13 inside it. The estimator is most biased exactly where the criterion has already rejected the bin, so the bias moves the edge of the band without changing which side a record falls on.

IV. Physical Reading of the Bound

Section III gives the depth of a collapse; a designer asks a different question—which bin should I use, and what do I change when none is usable? Four consequences of (5) answer it, and they bind in this order. *First*, the failure set is a band rather than a direction, so resolution becomes a placement problem. *Second*, the swing is a switch that normal breathing

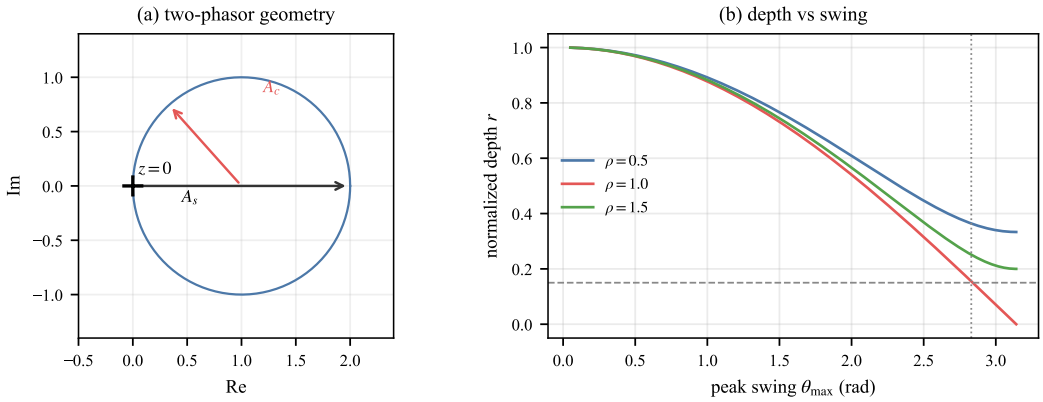


Fig. 1. Two-phasor geometry and depth bound. (a) The trajectory of $z = A_s + A_c e^{j\theta}$ is a circle of radius A_c centered at A_s; the origin is reached only when $A_c = A_s$ and $\theta = \pi$. (b) Normalized depth r versus peak swing $\theta_{\max}$ for three amplitude ratios; dashed line: $r = 0.15$; dotted line: $\theta_{\max} = 2.84$ rad.

has already thrown, so it is not a second knob. *Third*, a null carries a phase only relative to a frame that lasts as long as the static phasor holds still. *Fourth*, noise keeps the depth away from zero, so every criterion reads against a floor.

A. The Failure Set Is a Band, Not a Direction

For a fixed swing, (4) has a single minimum at $\rho = 1$ and weakens on both sides, so neither a finer nor a coarser bin is uniformly safer: the failure set is a band rather than a half-space. A fine bin isolates the chest and leaves the static return dominant ($\rho \ll 1$)—the trajectory becomes a small circle far from the origin, nulls are suppressed, and part of the displacement signature goes with them. A coarse bin sums the chest with the room and pulls ρ toward one, where the trajectory reaches the origin. What makes this usable is that the band has a closed form: once the swing covers the antiphase direction, (5) makes the 15% criterion exactly $\rho \in [0.7391, 1.3529]$, independent of how hard the subject breathes.

B. The Swing Is a Switch Already Thrown

This is the consequence a designer acts on. Normal breathing at 60 GHz puts the swing past π in 76.4% of the records measured here (Fig. 2b), so for most of the operating envelope the swing is not a free parameter at all: it has already saturated, and (5) leaves a single quantity to design against. That quantity is the amplitude ratio, and the amplitude ratio is set by how many scatterers share a bin, i.e. by the range resolution. In this regime a chirp bandwidth is not a resolution setting that happens to correlate with signal quality: it is what moves the amplitude ratio, and the ratio is the variable the criterion is written in.

C. Cancellation or Amplitude Modulation

The null has an alternative explanation: if the chest motion fades, the envelope dips just as deeply. At a geometric null the motion phasor must point opposite the static phasor:

$$\kappa = \cos \angle(a(t), -A_s(t)) \approx 1, \qquad a(t) = z(t) - A_s(t). \quad (10)$$

The two are told apart by κ, which approaches one only when the dip is a cancellation; we therefore treat it as a manipulation check, and only high-alignment events count as geometric nulls.

D. The Reference Frame of a Null

The null direction is $\angle(-A_s(t))$, not a fixed axis in the complex plane. If A_s is constant, folding null instants onto the respiratory phase is well defined; if A_s rotates, the null direction rotates with it and a fixed-frame phase loses meaning. The quantity that survives is the angle of the motion phasor relative to the instantaneous null direction,

$$\psi(t) = \angle a(t) - \angle(-A_s(t)) \pmod{2\pi}. \quad (11)$$

A null is an event with $\psi \approx 0$; the respiratory phase at which it happens is $\angle a(t)$ shifted by the rotation of A_s. Two consequences follow. First, a fixed-frame analysis is the special case $\angle A_s = \text{const}$, so its validity is a property of the model rather than an empirical assumption. Second, the usable length of a fixed-phase analysis is set by the accumulated rotation $\Delta\varphi_s = |\dot{\varphi}_s| T$ over the window T: once $\Delta\varphi_s$ becomes comparable to the phase resolution of the folding histogram, the null instants smear and the Rayleigh statistic $\bar{R}$ drops even though the physics is unchanged. This is what the drift-window sweep measures in Section VI-D: $\bar{R}$ is flat for 5–15 s windows and degrades at 20 s, bounding the usable $\Delta\varphi_s$ in these recordings.

E. Why the Depth Never Reaches Zero

Noise puts a floor under the depth, and that floor decides how much static-to-noise ratio a bin needs. Adding w gives $z = A_s + A_c e^{j\theta} + w$. At the null instant the deterministic part vanishes, so the observed minimum is the noise sample itself, $|z|_{\min} = |w|$, projected on the null direction. With complex Gaussian noise of per-component standard deviation σ, the typical depth floor is therefore

$$r_{\min} \simeq \frac{\mathrm{E}|w|}{A_s + A_c} = \frac{\sqrt{\pi/2}\,\sigma}{2A_s} \quad (\rho = 1), \quad (12)$$

so $r = 0$ is unobservable in any finite-SNR bin and the depth criterion reads against a floor.

Requiring $r_{\min} < 0.15$ in (12) gives $A_s/\sigma \approx 4.2$, about 12.4 dB of static-to-noise amplitude ratio in the bin. That is the single-sample value. A detector takes the minimum over N frames, and the minimum of N samples lies below the single-sample mean: Monte Carlo gives $\mathrm{E}[\min_n |w|]/\mathrm{E}|w| = 0.071$ at $N = 100$ and 0.020 at 1200, so in a 30 s window the same noise reaches well below 15% on its own. Three quantities therefore stay apart: the single-frame noise amplitude, the minimum over N frames that decides whether a window can manufacture a dip, and the detector threshold $r_{\mathrm{th}} = 0.3$ on the record median. This is why the gate is calibrated on data rather than read off (12), and why a deep null in a strong-clutter bin is optically shallower than the geometry alone predicts.

V. From Model to Decision

The model says which bins *can* fail; a working system needs to know which one *will* fail before it is used. Two named parts close that gap. The *gate* is a dip fraction D, measurable in 20–30 s of pre-lock data against a threshold this section calibrates. The *design rules* R1–R4 turn the band of Section IV into a placement decision. A scoring penalty then applies the gate to bin ranking.

A. Dip-Fraction Criterion

The first quantity needs neither phase nor model: it is the fraction of frames in which the bin is deep. For a window of N frames,

$$D = \frac{1}{N} \sum_n \mathbf{1} \left\{ |z[n]| < r_{\mathrm{th}} \operatorname{median} |z| \right\}, \qquad r_{\mathrm{th}} = 0.3. \tag{13}$$

D is an early-warning statistic: it rises before the hard-failure threshold $r < 0.15$, and its threshold is set by the grid (Section V-D). In the 60 GHz three-bandwidth dataset, records containing at least five deep nulls have $D = 3.8$–5.9% in the selected bin against 0–0.25% in the remaining candidate bins (Wilcoxon $p \le 1.6 \times 10^{-2}$ per bandwidth), and Section VI-A shows that the statistic also predicts the second half of a record from its first.

B. The r–D Bridge

What makes that fraction interpretable is a closed-form link to the depth bound. For a sinusoidal phase $\theta(t) = \theta_{\max} \cos \omega t$ and $\rho = 1$, $|z|/|z|_{\max} = |\cos(\theta/2)|$. The fraction of time with $|z|/|z|_{\max} < r$ is

$$D_{\max}(r; \theta_{\max}) = \frac{2}{\pi} \arccos\left(\frac{2 \arccos r}{\theta_{\max}} \right), \tag{14}$$

provided the swing is deep enough for the arc to spend time below r, that is $2 \arccos r < \theta_{\max}$; for a shallower swing the fraction is zero and the bridge does not apply. Because (13) is normalized by the median, the effective threshold is $r_{\mathrm{eff}} = r_{\mathrm{th}} \operatorname{median} |z|/|z|_{\max}$; (14) is evaluated at r_{eff}. For $\theta_{\max} \approx 2.85$ rad and $r_{\mathrm{eff}} \approx 0.15$, (14) gives $D \approx 5\%$, consistent with the grid calibration and with the measured 3.8–5.9%.

C. Design Rules R1–R4

- R1 *Bin rejection.* Reject bins whose deep-null event count n_{deep} reaches 5 in a 30 s pre-lock window (3 in 20 s), counted raw. The pre-lock window and the running monitor have different jobs: the first builds the ranking, the second re-counts and re-ranks when the count moves. The window length is a delay-performance trade, and it is flat between 20 and 30 s (Section VI-A). This reproduces the operating point of $D > 1\%$ (within-record sensitivity 0.56 vs 0.57, specificity 0.99 vs 0.98); regressing n_{deep} on $|A_s|/R$ is deliberately *not* applied, because the ratio is the mechanism rather than a confounder (see Sec. VI).
- R2 *Ratio avoidance.* Prefer bins whose estimated ρ is away from one.
- R3 *Distance.* Choose a range where the estimated ρ leaves the failure band.
- R4 *Breathing-depth exemption.* A peak-to-peak displacement below 2.26 mm at 60 GHz keeps $\theta_{\max} < 2.84$ rad, so the 15% depth cannot be reached at $\rho = 1$ and the bin is exempt. The number follows from (6) and scales with λ. Applies to 20.5% of our records; it is the only rule whose scope is set by the subject rather than by the room.

D. Numerical Grid Calibration

A 143-combination grid sweeps ρ and $\theta_{\max}$ with additive noise (three seeds per point). The failure set defined by $r < 0.15$ has the bounding box $\rho \in [0.98, 1.15]$, $\theta_{\max} \ge 2.83$ rad. The swing bound agrees with the analytic $2 \arccos 0.15 = 2.840$ rad to 0.01 rad, an independent check that the grid resolves the mechanism rather than the noise realization. The ρ interval is a bounding box rather than a level set: it is the grid's resolution limit at swings below π, and the analytic set widens to $[0.7391, 1.3529]$ at π and then stays there. The grid only has to resolve the onset: above π the set is analytic, (5), which is why a 3.2 rad sweep calibrates a range running to 27.5 rad. Fig. 2(a) draws the set over the full measured range, where the 15% contour becomes vertical at π—the visual statement of the saturation. The 1% dip fraction separates grid points inside the failure region from those outside, with (14) interpolating between grid points; Fig. 2 shows the calibrated region.

E. Scoring Penalty

Ranking is where the statistic pays for itself: it is applied as a penalty linear in D until it saturates.

$$S = S_0 - w_D \min\left(\frac{D}{D_{\mathrm{ref}}}, 1 \right), \qquad D_{\mathrm{ref}} = 5\%, \quad w_D = 0.25. \tag{15}$$

The ramp in (15) is linear in D and saturates at 5%, the upper end of the 2–6% seen on null-bearing bins; 1% is the hard rejection threshold of rule R1, not the saturation point. D is zero at the *selected* bin in 93 of the 161 Lei records, yet nonzero in at least one candidate bin in 116 of them: selection already avoids null bins, so a zero at the winner does not mean the statistic is inert. The same pattern holds on the 20-record capsule benchmark, where D at the selected bin is nonzero in 7/20 records and above 1% in 3/20, while 26–73 of the 90

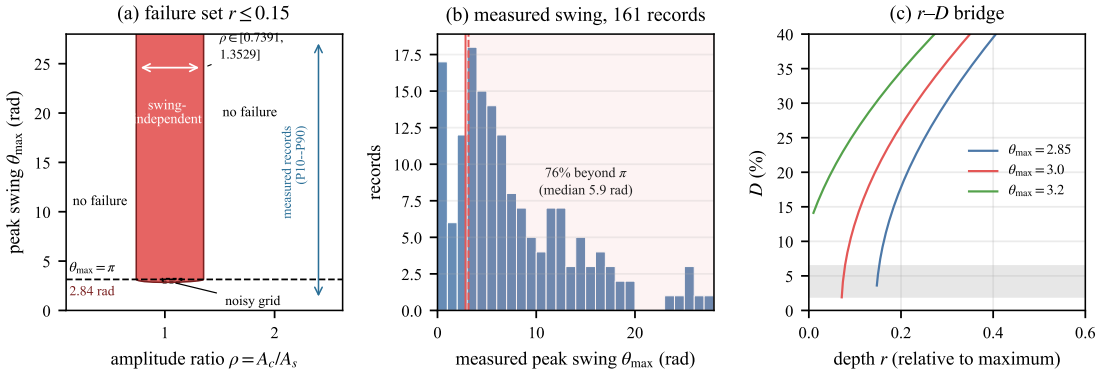


Fig. 2. Failure set and its calibration. (a) Analytic failure set $r \le 0.15$ over the full measured swing range; the set is a ρ band whose width no longer depends on $\theta_{\max}$ once the arc covers the antiphase direction ($\theta_{\max} \ge \pi$, dashed). Dashed box: the noisy 143-point grid, which stops at 3.2 rad. (b) Peak swing measured on the 161 records (median 5.92 rad; 76.4% beyond π, shaded). (c) The r–D bridge for three swings; shaded band: the few-percent range used by the grid.

(θ, bin) cells exceed 1% in the same 20 s window. The penalty therefore acts on the candidate set. It leaves 18/20 records bit-identical, improves one through an angle re-selection (heart-rate error 8.53→0.66 bpm), and regresses one by 0.20 bpm (aggregate mean absolute error 6.01→5.63). Only two paired differences are non-zero, so no significance claim is made (supplementary Table S2).

VI. Evidence

Each of the five claims below is turned into an experiment in the same way: we start from the equation or the deployment constraint that makes the claim testable, design the measurement that isolates that constraint, report what happens, and state what the observation establishes. Every count below is a record—161 of them (9 geometries × 6 repeats × 3 bandwidths, less one missing 7.5 cm record). Tests of bandwidth use position-level pairing ($n = 9$); event-level statistics use a record-clustered bootstrap; the held-out figure is leave-one-position-out. Table I collects the numbers; the dataset summary, the breath-hold condition decomposition, the window sweep, the capsule phase portrait and the bandwidth example are in the supplementary material.

A. Evidence 1: Resolution Moves the Failure Rate

The question. Equation (5) makes the criterion depend on one quantity, the amplitude ratio ρ, and ρ is set by how many scatterers share a range bin. Range bins coarsen as $\Delta R = c/2B$ grows, so the model predicts that chirp bandwidth—a parameter normally chosen for resolution alone—must move the failure rate. Testing that prediction requires a scene in which the people, the room and the geometry are fixed and only B changes.

The test. The Lei dataset [19] provides exactly that scene: two seated subjects recorded with a TI IWR6843ISK board and a DCA1000EVM card at 60 GHz (3 TX × 4 RX, 200 samples at 4 Msps, 1200 frames at 20 Hz, one chirp per frame), where sweeping the slope over 60/50/40 MHz/μs gives 3.0/2.5/2.0 GHz of bandwidth and 5.0/6.0/7.5 cm bins without moving anyone. The same nine geometries are used at every bandwidth, each repeated six times, which is what turns a comparison across recordings into a controlled one. For each record we take the 12-channel complex mean of the range profile and select the bin with the largest standard deviation in the 0.1–0.6 Hz respiratory band within $\lfloor (r-0.25)/\Delta R \rfloor$ to $\lfloor (r+0.05)/\Delta R \rfloor$ of each known subject range, so that the fixed reflector near 1.4 m is never selected. A deep null is a local minimum of $|z|$ below $0.35\times$ the record median, separated from the previous minimum by 0.25 s, with prominence 0.15σ of the envelope; D uses $0.3\times$ median and counts frames, and n_{deep} counts events.

What happens. Fig. 3(a) shows the fraction of records carrying at least five deep nulls rising from 13.0% at 5.0 cm to 48.1% at 6.0 cm and 62.3% at 7.5 cm (Table I); what the model predicts is a step, not a monotone trend, because the band is crossed rather than approached. The 5.0 to 6.0 cm step is significant at the record level (Fisher $p = 1.3\times10^{-4}$) and at the position level (paired $p = 0.031$ over the nine geometries, which does not treat repeats as independent), and across four depth thresholds, three prominence levels and three event cut-offs the ordering holds in all 36 combinations. The ratio itself moves the way the model requires: $|A_s|/R$ falls from 19.7 to 4.65 and 2.57, because a fine bin isolates the chest's own static return while a coarse bin sums the chest with room statics. Within each bandwidth the null count decreases with the distance of $|A_s|/R$ from one (Spearman -0.63, -0.81 and -0.56), and 42% of null-bearing records have $|A_s|/R < 2$ against 5% of the rest. The band has two safe sides, which is why 38% of 7.5 cm and 52% of 6.0 cm records carry no null at all, and the record-level ratio is a conservative estimate of the instantaneous one because R is a respiratory-band RMS.

What it establishes. Within the radar, scene and bandwidth range studied, bandwidth is not a resolution setting that happens to correlate with signal quality: it changes how the returns add up, and with that the failure rate. The same conclusion holds event by event, which is the stronger form—for each of the 1450 detected dips a circle fitted to the ±2 s window gives a local ratio whose median is 0.87, 0.77 and 0.98, on the band

TABLE I
RANGE-RESOLUTION CONTROL

Bandwidth	Bin	Records	Records with ≥5 nulls	Median nulls	D null bins	D other bins	κ	$\bar{R}$	$\lvert A_s\rvert/R$
3.0 GHz	5.0 cm	54	13.0% (7)	0	5.9%	0.0%	0.85	0.67	19.7
2.5 GHz	6.0 cm	54	48.1% (26)	3	3.8%	0.0%	0.89	0.74	4.65
2.0 GHz	7.5 cm	53	62.3% (33)	11	4.2%	0.25%	0.83	0.65	2.57

D and the alignment statistics are computed within the null-bearing records; $|A_s|/R$ is the median over all records.

where (5) predicts a deep null; predicted and observed depths agree to a median absolute error of 0.057, and the local ratio falls inside $[0.7391, 1.3529]$ at 51.8% of dips against 15.4% of random instants, an enrichment of $3.4\times$.

Is it predictable before it is needed? A gate is only useful if it fires early, so we split every record into two 30 s halves and asked whether the first predicts the second over all 1417 candidate bins. It does: pooled area under the curve 0.92 for $n_{\mathrm{deep}} \geq 1$ (base rate 27%), within-record median 1.00, with $n_{\mathrm{deep}} \geq 3$ per 30 s giving sensitivity 0.70 and specificity 0.96 and ≥ 5 giving 0.56 and 0.99; on the 161 selected bins the depth statistic D reaches 0.91. Calibrating by position, leave-one-position-out over the nine geometries, gives 0.89 held out against 0.92 in-sample, with the Youden threshold equal to one event per 30 s in all nine folds. The pre-lock length is a delay-performance trade rather than a free parameter: the same test gives 0.836 at 10 s, 0.891 at 20 s, 0.937 at 30 s and 0.919 at 45 s, so 20–30 s sits on a plateau and 10 s does not. Residualizing on $|A_s|/R$ is counterproductive, because the ratio is the causal driver rather than a confounder: removing it drops the within-record area under the curve from 1.00 to 0.73.

Is the count doing the work, or would any envelope statistic do? On the same 161 records and 30 s pre-lock, three alternatives predict the second half: envelope variance (AUC 0.947), the envelope's fifth percentile (0.963) and the dip fraction D (0.906), against 0.937 for n_{deep}, on the weak label of any dip; on the label the gate is built to predict—five or more—they give 0.926, 0.912 and 0.903 against 0.934. The envelope statistics match the count on the weak label and fall behind on the strong one: one dip is not a failure, five are. What the area under the curve measures is the persistence of a bin's state across the two halves of a record, not performance on a new subject in a new room. The dip frequency also matches the respiratory peak (median relative error 0.00 at 6.0 cm) and the synchronized ECG places the subject at rest, so the selected bin carries a person's breathing rather than a static object.

B. Evidence 2: Separating Cancellation from an Amplitude Fade

The question. A dip in $|z|$ is not by itself proof of cancellation: if the chest motion fades, the envelope dips too, and the two mechanisms leave the same depth statistics. Equation (10) makes the distinction testable—a geometric null requires the motion phasor to point opposite the static one—and it also implies a shape: to go deeper, the trajectory must pass closer to the origin, so the phase turn across the dip must grow with depth. An amplitude fade has no such coupling. The experiment therefore has to test both the alignment and the depth coupling.

The test. We measure the alignment $\kappa = \cos\angle(a(t), -A_s(t))$ on every null-bearing record, and we measure the phase step across a ± 0.2 s window at every dip. Because any detector would report dips in either world, we then pass two synthetic worlds through the identical detector and compare: a two-phasor world ($\rho \in [0.85, 1.15]$, swings 4–12 rad) and an amplitude-modulated world whose swing stays below π, so the arc never reaches antiphase, and whose motion amplitude falls to 5% twice per cycle. The two worlds are matched in dip depth (median 0.88 and 0.90 against 0.78 measured), so they differ only in mechanism.

What happens. Alignment: the median κ is 0.85, 0.89 and 0.83 at the three bandwidths, 86–92% of null-bearing records pass the $\kappa \geq 0.6$ gate (6/7, 24/26, 29/33), and the drift-frame Rayleigh statistic is 0.67, 0.74 and 0.65—measured under a harder condition than the 4.5 cm set, with 20 Hz frames and two occupants. Shape: of the 1450 detected dips, 1231 enter this analysis—12 have a window crossing a record edge, 18 a rim-to-floor difference below 0.02 of the median, and 189 a shape ratio outside $(0.02, 0.98)$, leaving both shoulders resolvable. Across that window the phase step grows with depth (medians 1.18, 1.47 and 2.34 rad over depths 0.65–0.8, 0.8–0.9 and above 0.9; Spearman $+0.33$, record-clustered bootstrap $[0.27, 0.39]$), and 20.7% of dips turn by more than 2.5 rad. The two-phasor world reproduces both ($+0.15$; 37.1% beyond 2.5 rad, 4699 dips), while the amplitude-modulated world reproduces neither (-0.01, $p = 0.72$; 0.8% beyond 2.5 rad, 998 dips).

What it establishes. The measured population sits between the two worlds but far closer to the geometric one: single dips separate with an area under the curve of 0.71, and dips that turn by more than 2.5 rad are 26 times more frequent than an amplitude fade can produce. The measured dips are consistent with cancellation rather than with a fading amplitude, and the gate can tell the two apart instead of counting envelope minima (Fig. 4). The separation is at the level of a population, not of a single dip: at an area under the curve of 0.71 the shape statistic is a discriminator between mechanisms, not a per-event verdict, and a dip turning through 2 rad remains consistent with either.

Both statistics come from the residual $a = z - \hat{A}_s$, so their robustness to $\hat{A}_s$ is checked rather than assumed. Perturbing $\hat{A}_s$ in amplitude by up to 30% and in phase arbitrarily leaves the separation of the two worlds unchanged: AUC 0.925, 0.928, 0.925 and 0.921 at 0, 10, 20 and 30%. The geometric fit supplies A_s with a residual of 0.321 of the fitted radius, so a real trajectory is a circle to within a third and

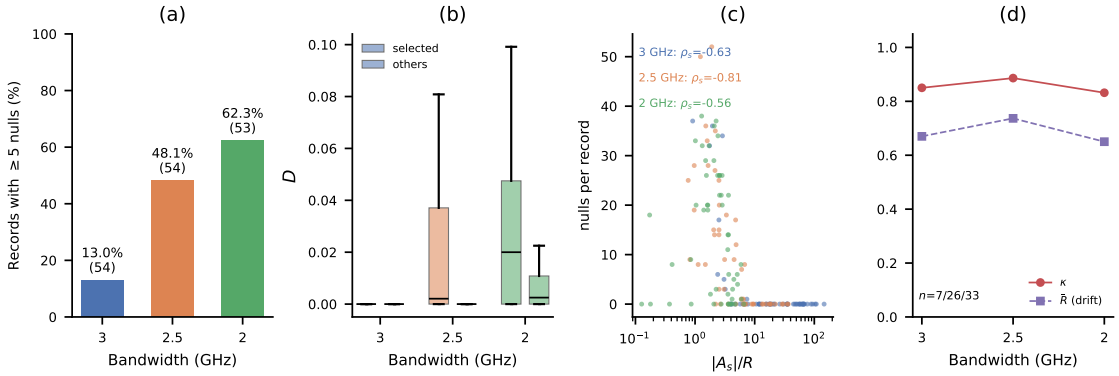


Fig. 3. Range-resolution control in the Lei dataset [19]. (a) Fraction of records with at least five deep nulls at 5.0, 6.0 and 7.5 cm ($n = 54/54/53$). (b) Dip fraction D for the selected bin and for the other candidate bins. (c) Null count versus $|A_s|/R$ (logarithmic); ρ_s is the Spearman correlation with $|\log(|A_s|/R)|$. (d) Median alignment κ and drift-frame Rayleigh $\bar{R}$ among null-bearing records ($n = 7/26/33$).

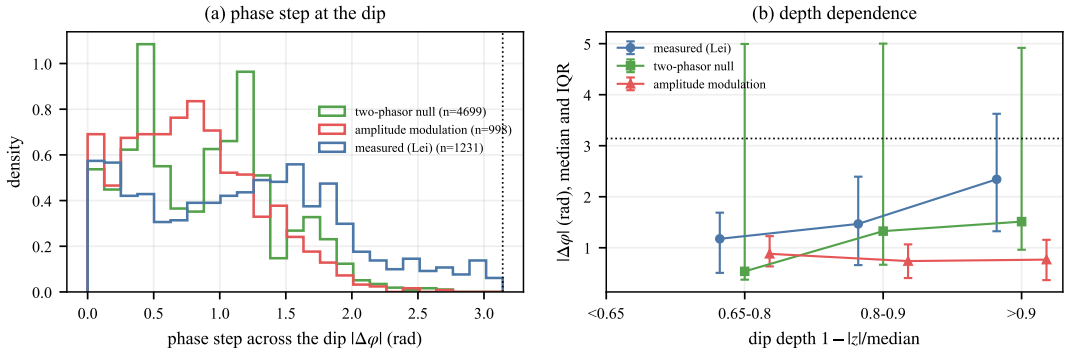


Fig. 4. Shape test separating a geometric null from an amplitude fade. (a) Phase step across the dip for the measured dips and for two synthetic worlds passed through the identical detector. (b) The same statistic against dip depth: a geometric collapse has to approach the origin to go deep, so the turn grows with depth; an amplitude fade has no such coupling. Dotted line: π.

not to within a percent—which is why the phase geometry serves as a discriminator between mechanisms rather than a reconstruction of the motion.

C. Evidence 3: Respiration Drives the Collapse

The question. The model requires a rotating phasor: a static bin cannot null by itself, whatever its amplitude. If that is right, removing the breath must remove the collapses—and a collapse that survives a breath-hold would falsify the model outright. The test therefore has to change respiration while holding gross motion fixed.

The test. 110 subjects × 4 conditions, 50 s at 10 Hz, 8 virtual channels × 64 bins at 480 MHz of sampled bandwidth and 31.25 cm bins, with 220 annotated breath-holds (median 11.0 s) [17]. At this resolution a bin sums the whole scene, which is exactly what makes the intervention clean: it moves respiration while the geometry stays put, and an accelerometer on the same time base controls for body motion.

What happens. Pooling both resting segments per subject gives 5.38 nulls/min inside the hold and 18.84/min outside; 80.9% of subjects are lower inside (Wilcoxon $p = 4.4 \times 10^{-12}$), and a Poisson model with subject-clustered errors gives a rate ratio of 0.438 (95% CI 0.348–0.551). Accelerometer motion is unchanged (median 0.047 versus 0.051, ratio 0.96).

What it establishes. The intervention removes 56% of the collapses with motion unchanged, which is what the model requires and what an estimator-side explanation cannot produce. The remaining 44% is a separate question that this test does not settle, and the criterion does not depend on settling it: it counts events in the bin that is actually used, whatever their origin.

D. Evidence 4: A Null Has a Phase, and the Frame Sets How Long It Holds

The question. The null direction is $\angle(-A_s(t))$, not a fixed axis, so (11) predicts that a null has a well-defined respiratory phase exactly as long as A_s is constant, and that the description degrades as A_s rotates—even though the physics is unchanged. That prediction is testable twice: the phase should cluster within a short record, and the clustering should fall away as the frame lengthens.

The test. Twenty 30 s records at 100 Hz with ECG R peaks [20], in which nine carry at least five deep dips; we fold the null instants onto the respiratory phase in a fixed frame and in the drift-compensated frame, and we repeat the folding over 5, 10, 15 and 20 s windows.

What happens. The median alignment is 0.91, 7/9 pass the geometric gate, and all seven lock to a fixed respiratory phase; the representative record (patient_15: $n = 18$, $\bar{R} = 0.83$, Rayleigh $p = 3.7 \times 10^{-6}$) concentrates its null instants at a preferred phase. Windows of 5, 10 and 15 s all give the same 7/9 locked, and the clustering degrades at 20 s.

What it establishes. The phase of a null is real but frame-dependent, and the usable frame in these recordings is bounded at about 20 s—which is a property of the accumulated rotation of A_s, not of the mechanism.

E. Evidence 5: The Rule Transfers to a Second Front End

The question. The thresholds were calibrated on one front end at 20 Hz. If the ordering depends on that calibration, the rule is a property of the dataset; if it survives a different front end, frame rate and bin width without retuning, it is a property of the mechanism.

The test. The emotion dataset [18] uses a different 60 GHz front end: five subjects, 38 trials, 78 s at 100 Hz, 4.5 cm bins. We apply the gate and the first-half/second-half split unchanged, with no parameter re-estimated.

What happens. The dip frequency matches the respiratory peak within 20% in 35/38 trials; 25/37 eligible trials (68%) pass the gate with a median alignment of 0.68; a fixed phase locks 13/25. The first half ranks the second with a Spearman correlation of $+0.58$ ($p = 1.5 \times 10^{-4}$; median-split area under the curve 0.74). Because 36 of 38 trials contain nulls in both halves, the informative form of this check is the rank statistic rather than a sensitivity/specificity pair.

What it establishes. The ordering survives the change of front end, frame rate and bin width with no recalibration. The gate also refuses the right record: P02 has a median alignment of 0.45, the lowest static-to-motion ratio in the pool and the fastest-rotating static phasor (14.7°/s), only 2 of its 6 trials pass, and the shape test assigns its dips to amplitude modulation.

F. What the Five Claims Jointly Establish

Respiration drives the collapse; resolution sets which ratios are available, and the ratio decides whether a null is possible; the dips are attributable to cancellation rather than fading; the closed form holds event by event; and the rule transfers without recalibration. The five answers are consistent with one mechanism: a respiratory-driven collapse governed by the amplitude ratio of a single range bin.

VII. Discussion

Design takeaways. (i) *Bandwidth sets the reachable range of ρ.* Halving B changes whether a null is possible at all; with the swing already saturated in four fifths of our records, ρ is the only quantity left to place, and since it is not single-valued in B the rule is to estimate it and place the bin. (ii) *The failure set is a band*, so the 13% to 62% progression is a step across it rather than a monotone trend in resolution. (iii) *The criterion is cheap and checkable in situ*: one median and one threshold per bin over 20–30 s, and at runtime the R1 event count n_{deep}, which needs no median.

What the penalty buys, and what it does not. The first claim is established: the statistic identifies bins at risk, and the penalty acts on the candidate set—26–73 of the 90 (θ, bin) cells exceed 1% in every record. The second is preliminary. On 20 capsule records it leaves the winning bin unchanged in 18/20, re-selects the angle in one (heart-rate error 8.53→0.66 bpm) and costs 0.20 bpm in one; beat-level F1 moves 0.536→0.532 and IBI RMSE 228→237 ms, a small beat-level trade for avoiding a collapsing bin. With two non-zero paired differences no significance is claimed. Identifying a dangerous bin and improving end-to-end heart rate are separate results, and the first is the one the evidence carries.

Why calibration alone does not close it. The natural repair is to estimate the static phasor and subtract it, as geometric circle fitting does. On these records that repair makes the failure commoner: removing the fitted centre raises the record-level null rate from 39.8% to 65.2% at the same 15% criterion, and all four alternative estimators do worse (supplementary Table S5). Depth is a ratio, so subtracting a centre changes the denominator. A record whose dip reaches only about 5% of the full envelope crosses the 30% line once the envelope is reduced to the residual sinusoid: the same swing still sweeps past the residual static term, and a shallow interference becomes a deep collapse. What decides survival is not how well the static term is removed but how close the residual ratio sits to one—which (5) bounds and the bandwidth places.

VIII. Scope

What the model assumes, and what would extend it. Each bin is described by one dominant static phasor and one motion phasor, and the three bin widths measured here all meet that premise—a chest and a wall can share a bin, the condition the model is for. In two-occupant records the selected bin is validated physiologically, and spatially where the occupants separate in range. Sub-centimetre raw ADC with known positions would let the ratio be computed from geometry rather than estimated, and would localize the two paths separately; that is the next measurement this line calls for.

What a third phasor changes. Adding static paths does not move the failure band: two static phasors behave as their sum, and the band stays at $\rho_{\mathrm{eff}} \in [0.7391, 1.3529]$ with its peak at 1.00. A second independent motion phasor suppresses it instead: two motions rarely reach antiphase together, and the dip fraction falls from 0.126 to 0.07 whatever their frequency relation. The same signature orders the four placements. With both occupants in one bin the median dip count is 0 ($n = 45$), with shared range but distinct bearing 2 ($n = 31$), with distinct range but shared bearing 8 ($n = 32$), and with both distinct 22 ($n = 30$); Kruskal–Wallis gives $H = 32.5$, $p < 10^{-4}$, while $|A_s|/R$ runs 0.99, 0.92, 0.89 and 0.83 in the same order, so proximity to $\rho = 1$ does not explain it. The criterion fails safe there, protecting exactly the bins in which a single motion dominates—the single-occupant chest this work targets.

Where the criterion applies. The criterion needs an estimable ratio, hence a bin that resolves the chest rather than the whole scene; at 31.25 cm it is not identifiable and the gate

abstains, so that dataset carries the intervention (Evidence 3), not the geometry. Where it is estimable, the band of (5) is the envelope and R1–R4 are its rules. At 20 Hz the record resolves the respiratory phase, the scale on which a null occurs and the gate counts; cardiac timing is a separate problem on which nothing here depends. The statistic also inherits the front end: quadrature imbalance turns the circle into an ellipse, so $|z|$ is stretched along one axis and D shifts with it, and a chain with appreciable imbalance should measure D after calibration.

IX. Conclusion

A range bin collapses when a static return and a chest return rotate into antiphase, and that event has a closed form. Two results carry the paper. The first is that the depth bound *saturates*: once the respiratory swing passes half a turn—which normal breathing at 60 GHz does in 76.4% of the records measured here—the criterion collapses to a band in the amplitude ratio alone, $\rho \in [0.7391, 1.3529]$. What a designer controls is therefore the amplitude ratio, and the amplitude ratio is the bin width.

The second turns resolution into a design variable. With the same people in the same nine geometries, the share of records carrying deep nulls runs from 13.0% at 5.0 cm to 48.1% at 6.0 cm and 62.3% at 7.5 cm, following the distance of the measured ratio from one.

The evidence closes from both sides. Breath-hold in 110 subjects removes 56% of the collapses with accelerometer motion unchanged, establishing the respiratory origin; the phase step across the 1231 dips that pass the shape filter grows with dip depth as a trajectory through the origin requires, separating the two mechanisms at population level; and the closed form predicts each of the 1450 detected dips to 0.057 in normalized depth. A gate built on these results—one count, one threshold—predicts the second half of a record from the first (pooled AUC 0.91 over 161 records, 0.89 held out by position) and keeps its ordering on a second front end with no recalibration (Spearman $+0.58$).

What a system should do with this fits in a sentence: measure the amplitude ratio, place the bin width so that the ratio leaves the band, and gate with one count. The null is not a dropout to be averaged away; it is an event with an address, and the address is computable before the bin is used.

Data and Code Availability

The four datasets are on Mendeley Data [19], Code Ocean [20], and Zenodo records 18599983 [17] and 16900785 [18]. Scripts and the command lines that regenerate every table and figure are available from the corresponding author.

Acknowledgment

The technical content and numerical results are the author's own. A large language model translated and edited the English text, altering no measurement, figure or conclusion; the author has verified the full text and takes responsibility for it.

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